UAV-Relaying Strategies for Free-space Quantum Key Distribution Systems

— Research Progress Report —

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Outline

I. Introduction
   - Motivation
   - QKD Protocols

II. Research Update
   - System Models
   - Results
   - Future research plan

Motivation: UAV-based QKD

Necessity of Quantum Key Distribution (QKD)

- Risk to Current Security: Conventional encryption (like RSA) could be broken by future quantum computers.
- QKD provides "Information-Theoretic Security" based on the quantum mechanics, which cannot be broken by any computer.

Necessity of UAV Integration

- No Cables Needed: In disasters or emergencies, fiber-optic cables are often broken or unavailable.
- Mobility & Flexibility: UAVs can fly to any location and act as mobile relay nodes to build a secure network.

Motivation: UAV-based QKD

Goal of this Research

Find the best communication method using QKD and multiple UAVs.
(→Find a scenario that minimizes trusted nodes but maintains high SKR.)

How can we improve security?

→ Reduce the number of trusted nodes.

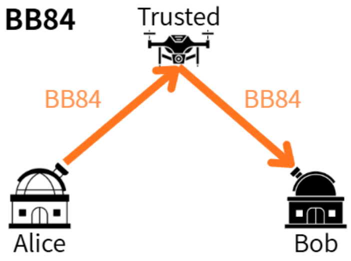
What is the Trusted Node?

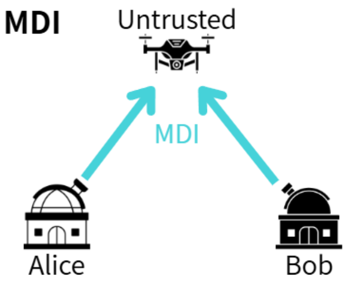
Nodes that hold key information → Nodes that must be trusted (High risk)

Protocol Features

BB84: All nodes are trusted nodes

MDI: Part of the nodes are untrusted nodes





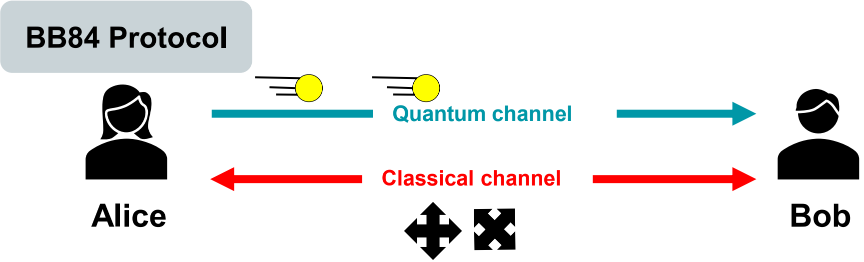
Quantum Key Distribution Protocol

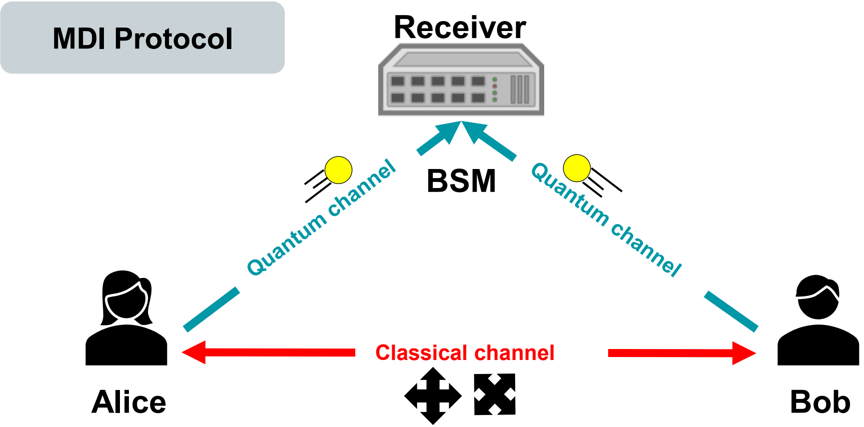
BB84 Protocol:

1. Alice sends single photons to Bob via a quantum channel.
2. They share measurement bases via a classical channel.
3. They keep the bits only when their bases match to form a secret key.

MDI Protocol:

1. Alice and Bob simultaneously transmit single photons to the UAV.
2. The UAV performs a Bell State Measurement (BSM) by interfering the two photons.
3. The UAV broadcasts the BSM results to Alice and Bob via a classical channel.
4. Alice and Bob generate the final secret key based on the results.





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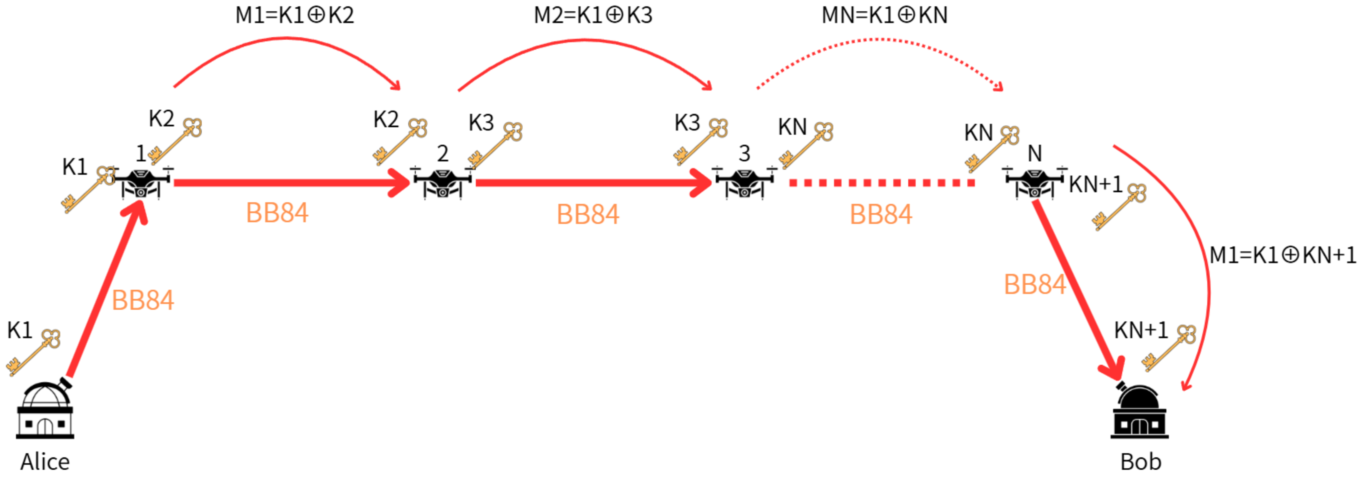
System Model (BB84 Multi-UAV)

Procedure for Key Sharing:

1. Generate keys for each link using the BB84 protocol.
2. UAVs sequentially encrypt and decrypt K1, relaying it to Bob via classical channels.
3. Alice and Bob successfully share a key K1.

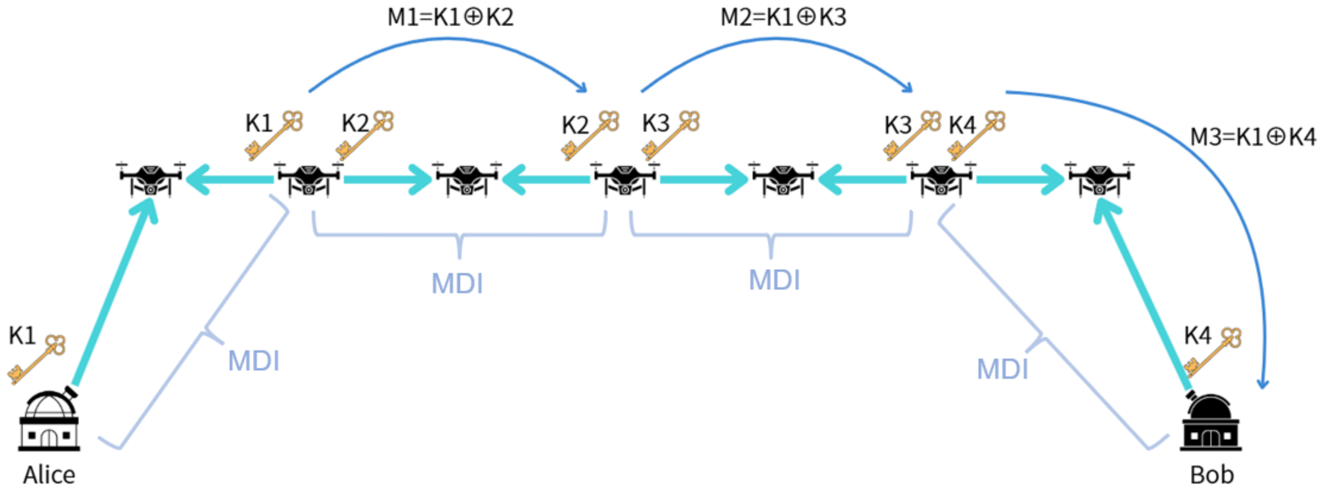


System Model (MDI Multi-UAV)

Case: Odd number of UAVs (N = 7)

Procedure for Key Sharing:

1. All links form perfect MDI pairs to generate keys.
2. UAVs relay the key K1 to Bob, similar to the All BB84 method.
3. Alice and Bob successfully share a key K1.

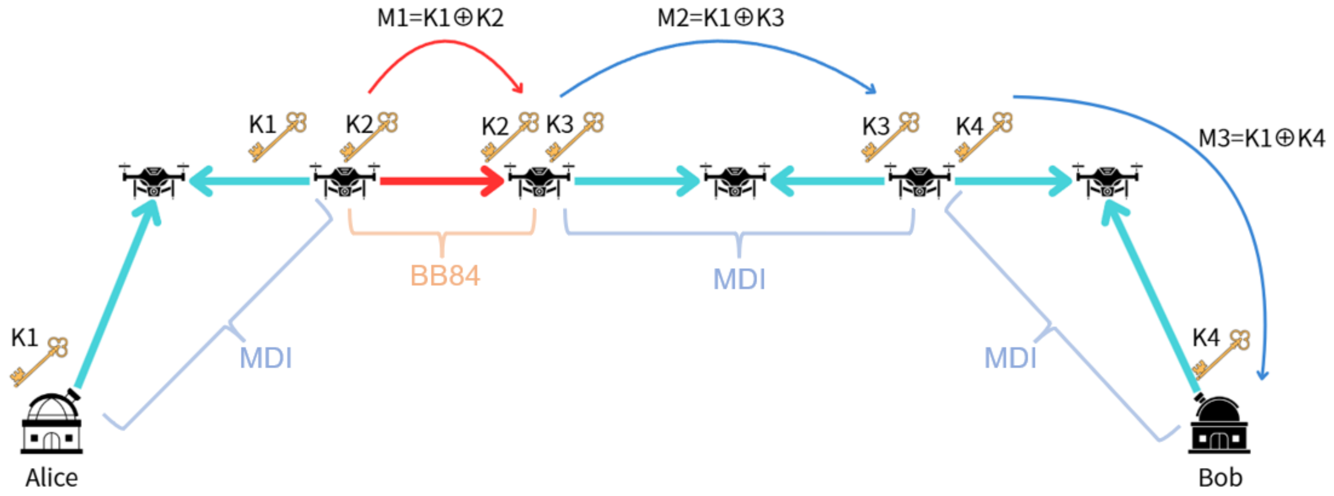


System Model (MDI Multi-UAV)

Case: Even number of UAVs (N = 6)

Procedure for Key Sharing:

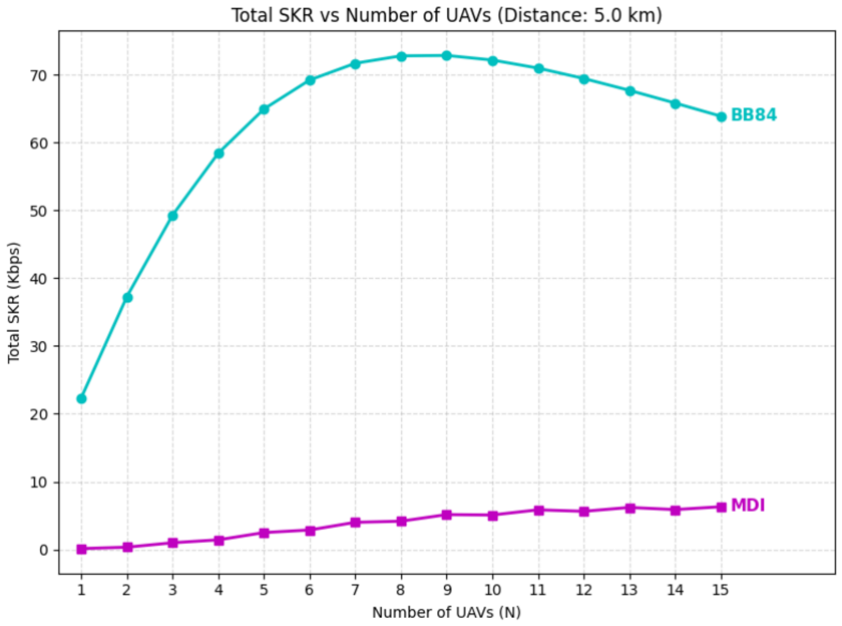
1. Generate keys using MDI pairs and our remining BB84 link.
2. UAVs relay the key K1 to Bob, similar to the All BB84 method.
3. Alice and Bob successfully share a key K1.



SKR vs. Number of UAVs (Fixed Distance: 5km)

Results:
- BB84>MDI: BB84 method achieves higher SKR.
- **Why MDI is lower:**
  MDI requires BSM across two links simultaneously, which naturally lowers the SKR.
- **Why SKR Increases (Left side of the peak):**
  As the number of UAVs increases, the distance of each link becomes shorter. This reduces atmospheric loss, leading to an initial rise in the SKR.
- **Why SKR Decreases (Right side of the peak):**
  However, adding too many UAVs increases the total number of hops. The accumulated QKD processing time at each node outweighs the benefits of shorter links, causing the overall SKR to decline.

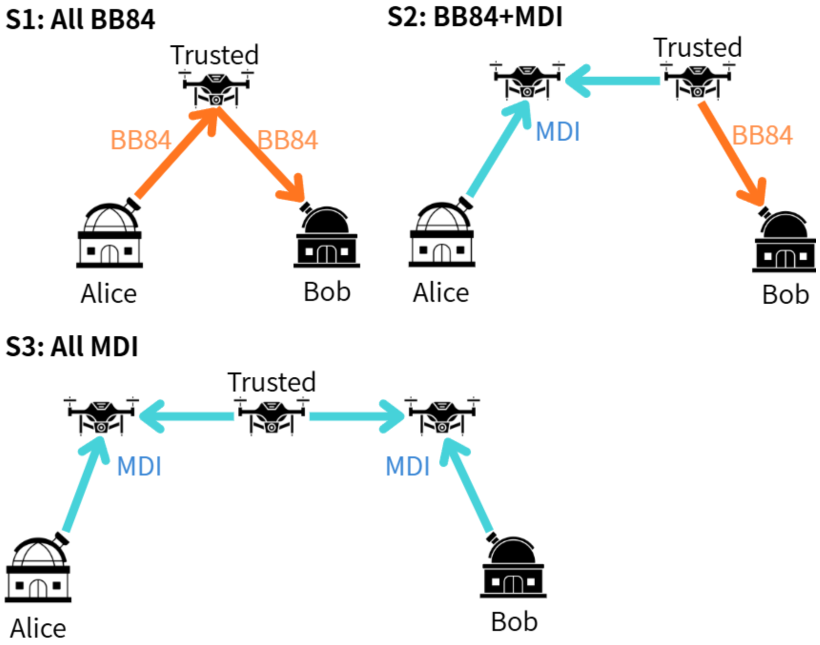
While the All BB84 method achieves a higher SKR, it requires every UAV to act as a fully Trusted Node. This increases network vulnerability.
→It is necessary to re-evaluate with a fixed number of Trusted Nodes.

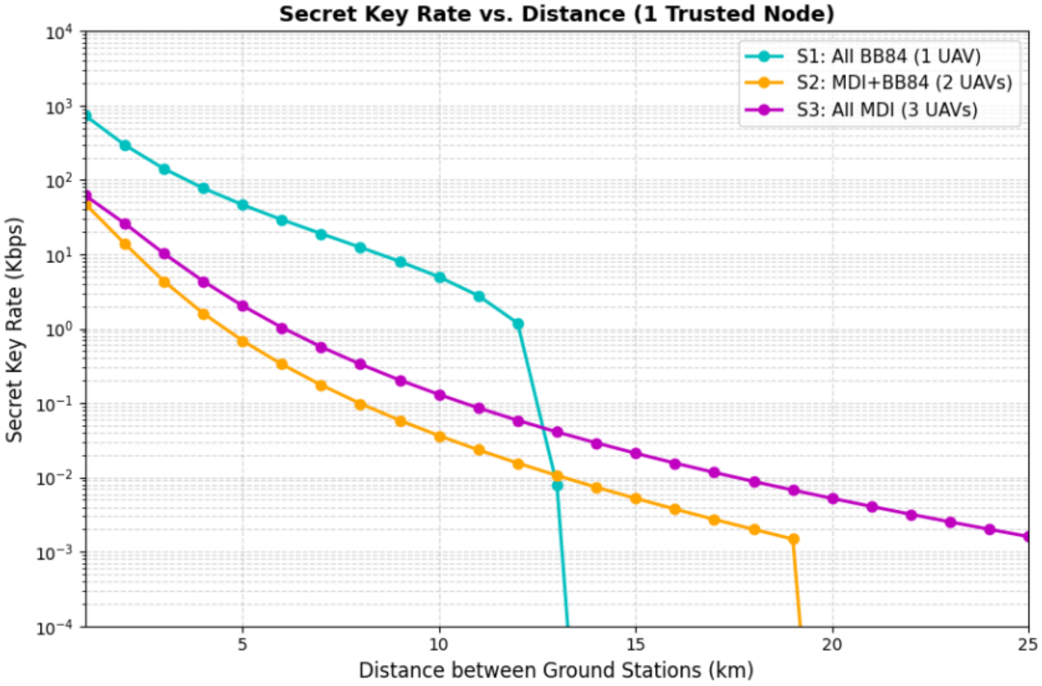


UAV Altitude: 500m
Distance between GS: 5km

SKR vs. Distance (1 Trusted Nodes)

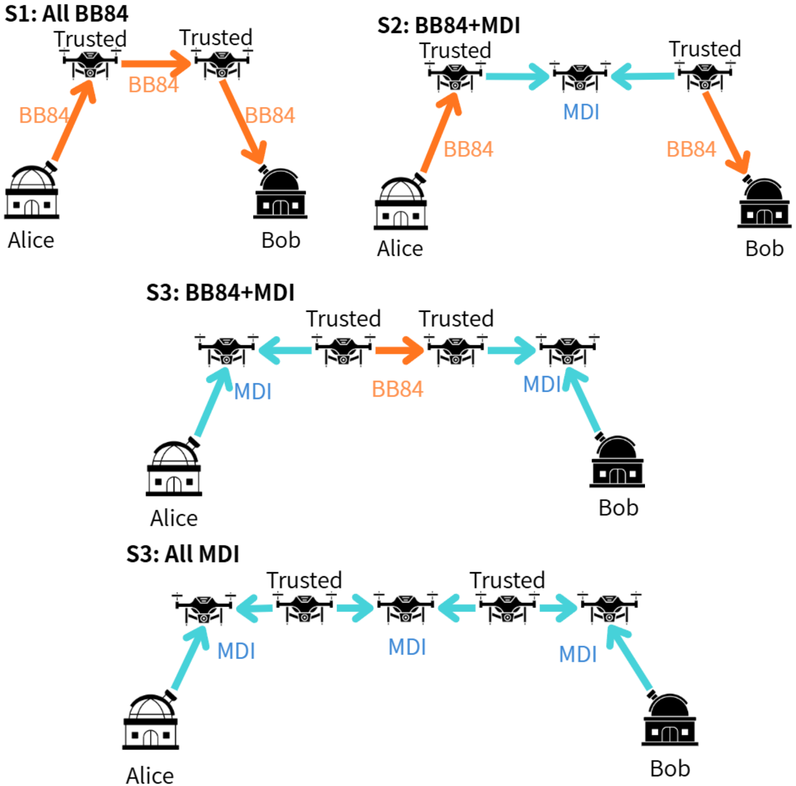
Scenarios:

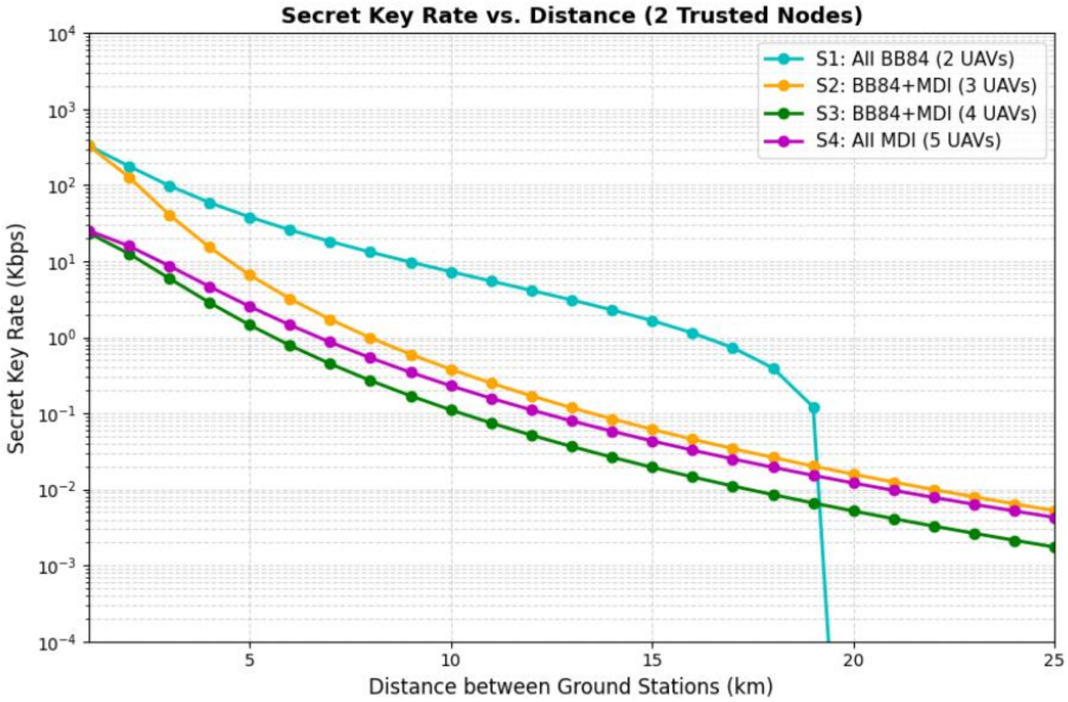




SKR vs. Distance (2 Trusted Nodes)

Scenarios:





Future Outlook & Plan

Goal of the Study

- Find the best communication method using QKD and multiple UAVs.

Conclusion from Today's Results

- Comparison of 3 conditions: number of UAVs, 1 Trusted Node, and 2 Trusted Nods.
- Therefore, the best scenario is different depending on the situation.

Future Plan

- Improve the simulation accuracy using Qiskit.
- Test more scenarios (e.g., change the number of Trusted Nodes).
- Find the best strategy: Minimize Trusted Nodes while maintain high SKR.

Thank you for listening